

Algebra Chapter 0 by Allufi

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June 15, 2026

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1 Preliminaries: Set Theory and Categories

1.1 Functions Between Sets

Problem 1

Prove that if $\sim$ is an equivalence relation on a set S , then the corresponding family of $\mathcal{P}_\sim$ define in sec 1.5 is indeed a partition of S : that is, its elements are nonempty, disjoint, and their union is S .

Proof. Suppose $\sim$ is an equivalence relation on a set S .

Let $[a] \in \mathcal{P}_\sim$. We have $a \in [a]$ by reflexivity, so $[a] \neq \emptyset$.

Let $a \in S$. We have $a \in [a] \in \mathcal{P}_\sim$, so $S \subseteq \bigcup \mathcal{P}_\sim$. Since every equivalence class is a subset of S , we have $\bigcup \mathcal{P}_\sim \subseteq S$. Thus $\bigcup \mathcal{P}_\sim = S$.

Suppose $[a], [b] \in \mathcal{P}_\sim$ with $[a] \cap [b] \neq \emptyset$, say $c \in [a] \cap [b]$. Then $a \sim c$ and $b \sim c$, so $a \sim b$ by symmetry and transitivity. For any $x \in [a]$, we have $x \sim a \sim b$, so $x \in [b]$. Thus $[a] \subseteq [b]$. By symmetry $[b] \subseteq [a]$, so $[a] = [b]$. ■

Problem 3

Given a partition $\mathcal{P}$ on a set S , show how to define a relation $\sim$ on S such that $\mathcal{P}$ is the corresponding partition.

Proof. We define $\sim$ by

$$a \sim b \iff \exists A, B \in \mathcal{P} \text{ such that } a \in A, b \in B, A \cap B \neq \emptyset.$$

■

Problem 5

Give an example of a relation that is reflexive and symmetric but not transitive. What happens if you try to use this relation to define a partition on the set?

Solution. Let $S = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (1, 3), (3, 1)\}$. This is reflexive and symmetric, but not transitive. Notice $(2, 1), (1, 3) \in R$ but $(2, 3) \notin R$.

If we attempt to define a partition via equivalence classes, the classes $[2] = \{1, 2\}$ and $[3] = \{1, 3\}$ overlap but are not equal, so the resulting family is not a partition of S . ■

Problem 6

Define a relation $\sim$ on the set $\mathbb{R}$ of real numbers by setting $a \sim b \iff b - a \in \mathbb{Z}$. Prove that this is an equivalence relation, and find a 'compelling' description for $\mathbb{R}/\sim$. Do the same for the relation $\approx$ on the plane $\mathbb{R} \times \mathbb{R}$ defined by declaring $(a_1, a_2) \approx (b_1, b_2) \iff b_1 - a_1 \in \mathbb{Z}$ and $b_2 - a_2 \in \mathbb{Z}$.

Proof. Let $x, y, z \in \mathbb{R}$. First, notice $x - x = 0 \in \mathbb{Z}$ so $x \sim x$. Next, suppose $x \sim y$ so $y - x \in \mathbb{Z}$. So $y - x = c$ for some $c \in \mathbb{Z}$. But then $x - y = -c \in \mathbb{Z}$ so $y \sim x$. Finally, suppose $x \sim y$ and $y \sim z$. So $y - x = c_1$ and $z - y = c_2$ for some $c_1, c_2 \in \mathbb{Z}$. Then $z - x = c_1 + c_2 \in \mathbb{Z}$. It follows that $x \sim z$ as required. ■

Proof. We now show $\approx$ is an equivalence relation on $\mathbb{R} \times \mathbb{R}$. Let $(x_1, x_2), (y_1, y_2), (z_1, z_2) \in \mathbb{R} \times \mathbb{R}$. First, notice $x_1 - x_1 = 0 \in \mathbb{Z}$ and $x_2 - x_2 = 0 \in \mathbb{Z}$ so $(x_1, x_2) \approx (x_1, x_2)$. Next, suppose $(x_1, x_2) \approx (y_1, y_2)$ so $y_1 - x_1 \in \mathbb{Z}$ and $y_2 - x_2 \in \mathbb{Z}$. So $y_1 - x_1 = c_1$ and $y_2 - x_2 = c_2$ for some $c_1, c_2 \in \mathbb{Z}$. But then $x_1 - y_1 = -c_1 \in \mathbb{Z}$ and $x_2 - y_2 = -c_2 \in \mathbb{Z}$ so $(y_1, y_2) \approx (x_1, x_2)$. Finally, suppose $(x_1, x_2) \approx (y_1, y_2)$ and $(y_1, y_2) \approx (z_1, z_2)$. So $y_1 - x_1 = c_1, y_2 - x_2 = c_2, z_1 - y_1 = d_1$, and $z_2 - y_2 = d_2$ for some $c_1, c_2, d_1, d_2 \in \mathbb{Z}$. Then $z_1 - x_1 = c_1 + d_1 \in \mathbb{Z}$ and $z_2 - x_2 = c_2 + d_2 \in \mathbb{Z}$. It follows that $(x_1, x_2) \approx (z_1, z_2)$ as required. ■

1.2 Functions Between Sets

Problem 1

How many different bijections are there between a set S with n elements and itself.

Proof.

$$n!.$$

Problem 2

Prove statement (2) of Proposition 2.1. You may assume that given a family of disjoint subsets of a set, there is a way to choose one element in each member of the family.

Theorem 1 (Proposition 2.1). Assume $A \neq \emptyset$, and let $f : A \rightarrow B$ be a function. Then f has a right inverse if and only if it is surjective.

Proof. Suppose f has a right inverse g . For $b \in B$, $f(g(b)) = (f \circ g)(b) = \text{id}_B(b) = b$, where $g(b) \in A$, so b is in the image of f . Thus f is surjective.

Conversely, suppose f is surjective. Since $A \neq \emptyset$, fix some $s \in A$. For each $b \in B$, since f is surjective there exists at least one $a \in A$ with $f(a) = b$; define $g(b)$ to be any such a . This is defined for all $b \in B$ since f is surjective. Then for all $b \in B$,

$$(f \circ g)(b) = f(g(b)) = b = \text{id}_B(b),$$

so $f \circ g = \text{id}_B$, meaning g is a right inverse of f . ■

Problem 4

Prove that 'isomorphism' is an equivalence relation (on any set of sets).

Proof. $\text{id}_A : A \rightarrow A$ is a bijection, so $A \cong A$.

Suppose $A \cong B$, so there exists a bijection $f : A \rightarrow B$. Since f is a bijection it has an inverse $f^{-1} : B \rightarrow A$, which is itself a bijection. Thus $B \cong A$.

Suppose $A \cong B$ and $B \cong C$, so there exist bijections $f : A \rightarrow B$ and $g : B \rightarrow C$. Then $g \circ f : A \rightarrow C$ is a bijection, since the composition of bijections is a bijection. Thus $A \cong C$. ■

Problem 5

Formulate a notion of epimorphism, in the style of the notion of monomorphism seen in sec 2.6, and prove a result analogous to Proposition 2.3, for epimorphisms and surjections.

Theorem 2. A function $f : A \rightarrow B$ is an epimorphism if for all sets Z and all functions $\alpha, \beta : B \rightarrow Z$,

$$\alpha \circ f = \beta \circ f \implies \alpha = \beta.$$

A function is surjective if and only if it is an epimorphism.

Proof. Suppose f is surjective. It follows that $f(A) = B$ so if $(\alpha \circ f)(A) = (\beta \circ f)(A)$ then $\alpha(B) = \beta(B)$ so $\alpha = \beta$.

Conversely, suppose f is not surjective. Let $b_0 \in B \setminus f(A)$. Let $\alpha, \beta : B \rightarrow \{0, 1\}$ be defined $\alpha = 0$ and $\beta(b) = 0$ for $b \neq b_0$, $\beta(b_0) = 1$. Then $\alpha \circ f = \beta \circ f$ but $\alpha \neq \beta$, so f is not an epimorphism. ■

Problem 9

Show that if $A' \cong A''$ and $B' \cong B''$, and further $A' \cap B' = \emptyset$ and $A'' \cap B'' = \emptyset$, then $A' \cup B' \cong A'' \cup B''$. Conclude that the operation $A \amalg B$ is well-defined up to isomorphism.

Proof. Let $f : A' \rightarrow A''$ and $g : B' \rightarrow B''$ be bijections. Now, consider the mapping $t : A' \cup B' \rightarrow A'' \cup B''$ defined by

$$t(x) = \begin{cases} f(x) & \text{if } x \in A', \\ g(x) & \text{if } x \in B'. \end{cases}$$

Let $x \in A' \cup B'$; since $A' \cap B' = \emptyset$, either $x \in A'$ or $x \in B'$, but not both. Thus $t(x)$ is a single well-defined element in $A'' \cup B''$, so t is a function.

To see that t is injective, suppose $t(x) = t(y)$. Since $A'' \cap B'' = \emptyset$, both $t(x)$ and $t(y)$ lie in the same subset, so either both $x, y \in A'$ (giving $f(x) = f(y)$, thus $x = y$ by injectivity of f) or both $x, y \in B'$ (giving $g(x) = g(y)$, thus $x = y$ by injectivity of g). Thus t is injective.

To see that t is surjective, let $z \in A'' \cup B''$. If $z \in A''$, then since f is surjective there exists $x \in A'$ with $f(x) = z$, thus $t(x) = z$. If $z \in B''$, then since g is surjective there exists $x \in B'$ with $g(x) = z$, thus $t(x) = z$. Thus t is surjective.

Since t is a bijection, $A' \cup B' \cong A'' \cup B''$.

Suppose $A \cong A'$ and $B \cong B'$ with $A \cap B = \emptyset$ and $A' \cap B' = \emptyset$. By the above, $A \cup B \cong A' \cup B'$ so $A \amalg B \cong A' \amalg B'$. Thus the disjoint union $A \amalg B$ is well-defined up to isomorphism. ■

Problem 10

Show that if A and B are finite sets, then $|B^A| = |B|^{|A|}$.

Proof. Suppose A and B are finite sets. Let $|A| = m$ and $|B| = n$. Then $|B^A| = |B_1 \times B_m| = |B_1| \cdot \dots \cdot |B_m| = n_1 \dots n_m = n^m = |B|^{|A|}$. ■

Problem 11

In view of Exercise 2.10, it is not unreasonable to use 2^A to denote the set of functions from an arbitrary set A to a set with 2 elements. Prove that there is a bijection between 2^A and the powerset of A .

Proof. Simply identify each element of $\mathcal{P}(A)$ with a binary string of length $|A|$, where the i -th bit is 1 if the i -th element of A is present and 0 if absent. This correspondence is clearly a bijection between $\mathcal{P}(A)$ and 2^A . ■

1.3 Categories

Problem 1

Let C be a category. Consider a structure C^{op} with

- $Obj(C^{op}) := Obj(C)$;
- for A, B objects of C^{op} (hence objects of C), $Hom_{C^{op}}(A, B) := Hom_C(B, A)$.

Show how to make this into a category.

Proof. We can use the $1_A \in Hom_C(A, A) = Hom_{C^{op}}(A, A)$ for all A as our identity morphism in C^{op} as well. We define composition in C^{op} as follows.

Given $f \in Hom_{C^{op}}(A, B)$ and $g \in Hom_{C^{op}}(B, C)$, by definition $f \in Hom_C(B, A)$ and $g \in Hom_C(C, B)$, so $f \circ g \in Hom_C(C, A) = Hom_{C^{op}}(A, C)$, and we set $g \circ^{op} f := f \circ g$.

The composition law is associative because it is associative in C . Clearly, the identities respect composition. Thus C^{op} is a category. ■

Problem 3

Formulate precisely what it means to say that 1_a is an identity with respect to composition in Example 3.3, and prove this assertion.

Proof. For all $a, b \in S$, 1_a is an identity if for all $f \in Hom(a, b)$ we have $1_b \circ f = f$ and $f \circ 1_a = f$. Let $f \in Hom(a, b)$, so $a \sim b$ and $f = (a, b)$. Since $\sim$ is reflexive, $1_a = (a, a)$ and $1_b = (b, b)$. Then $f \circ 1_a = (a, b) \circ (a, a) = (a, b) = f$ and $1_b \circ f = (b, b) \circ (a, b) = (a, b) = f$. That is, the identity respects composition from the left and right. ■

Problem 5

Explain in what sense Example 3.4 is an instance of the categories considered in Example 3.3

Proof. We take the elements $A \subset S$ to be objects in the category. Then $\subset$ corresponds to a relation $\sim$ which is reflexive and transitive. ■

Problem 6

Define a category V by taking $\text{Obj}(V) = \mathbb{N}$ and letting $\text{Hom}_V(n, m) =$ the set of $m \times n$ matrices with real entries, for all $n, m \in \mathbb{N}$. Use product of matrices to define composition. Does this category 'feel' familiar?

Proof. Let $M_{n,m} \in \text{Hom}_V(n, m)$. We take 1_n to be the $n \times n$ identity matrix I_n . Then notice

$$I_m M_{n,m} = M_{n,m} \quad \text{and} \quad M_{n,m} I_n = M_{n,m}.$$

Composition is defined by matrix multiplication. Given $M_{n,m} \in \text{Hom}_V(n, m)$ and $M_{m,k} \in \text{Hom}_V(m, k)$, their composite $M_{m,k} \circ M_{n,m} \in \text{Hom}_V(n, k)$ is well-defined since the dimensions are compatible. Associativity follows from associativity of matrix multiplication. This category feels like the category of finite-dimensional real vector spaces with linear maps, since a morphism $n \rightarrow m$ corresponds exactly to a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^m$. ■

Problem 7

Define carefully objects and morphisms in Example 3.7, and draw the diagram corresponding to composition.

Proof. Let C be a category and fix an object $A \in \text{Obj}(C)$. We will name our constructed category C_A . Now define $\text{Obj}(C_A)$ to be all morphisms from A to any object of C . Thus, an object of C_A is a morphism $f \in \text{Hom}_C(A, Z)$ for some object $Z \in C$. Pictorially, an object of C_A is an arrow $A \xrightarrow{f} Z$ which we can draw top-down

$$\begin{array}{c} A \\ \downarrow f \\ Z \end{array}$$

Now, let f_1, f_2 be objects of C_A , that is, two arrows

$$\begin{array}{ccc} A & & A \\ \downarrow f_1 & & \downarrow f_2 \\ Z_1 & & Z_2 \end{array}$$

in C . Morphisms $f_1 \rightarrow f_2$ are defined to be commutative diagrams

$$\begin{array}{ccc} & A & \\ f_1 \swarrow & & \searrow f_2 \\ Z_1 & \xrightarrow{\sigma} & Z_2 \end{array}$$

in the 'ambient' category C . That is, a morphism $f_1 \rightarrow f_2$ corresponds precisely to a morphism $\sigma : Z_1 \rightarrow Z_2$ in C such that $\sigma \circ f_1 = f_2$.

Given composable morphisms $f_1 \xrightarrow{\sigma} f_2 \xrightarrow{\tau} f_3$, that is, morphisms $\sigma : Z_1 \rightarrow Z_2$ and $\tau : Z_2 \rightarrow Z_3$ in C satisfying $\sigma \circ f_1 = f_2$ and $\tau \circ f_2 = f_3$, their composition corresponds to the diagram

$$\begin{array}{ccccc} Z_1 & \xrightarrow{\sigma} & Z_2 & \xrightarrow{\tau} & Z_3 \\ & \searrow f_1 & \downarrow f_2 & \swarrow f_3 & \\ & & A & & \end{array}$$

The composite morphism $f_1 \rightarrow f_3$ is then $\tau \circ \sigma : Z_1 \rightarrow Z_3$, corresponding to the commutative diagram

$$\begin{array}{ccc} Z_1 & \xrightarrow{\tau \circ \sigma} & Z_3 \\ & \searrow f_1 & \swarrow f_3 \\ & & A \end{array}$$

which commutes since $(\tau \circ \sigma) \circ f_1 = \tau \circ (\sigma \circ f_1) = \tau \circ f_2 = f_3$. ■

Problem 8

A subcategory C' of a category C consists of a collection of objects of C , with morphisms $\text{Hom}_{C'}(A, B) \subseteq \text{Hom}_C(A, B)$ for all objects A, B in $\text{Obj}(C')$, such that identities and compositions in C make C' into a category. A subcategory C' is full if $\text{Hom}_{C'}(A, B) = \text{Hom}_C(A, B)$ for all A, B in $\text{Obj}(C')$. Construct a category of infinite sets and explain how it may be viewed as a full subcategory of Set .

Proof. Let Set_∞ denote the category whose objects are all infinite sets, with morphisms

$$\text{Hom}_{\text{Set}_\infty}(A, B) = \text{Hom}_{\text{Set}}(A, B)$$

for all infinite sets A, B . Clearly, it's a full subcategory of Set . ■

Problem 9

An alternative to the notion of multiset introduced in §2.2 is obtained by considering sets endowed with equivalence relations; equivalent elements are taken to be multiple instances of elements 'of the same kind'. Define a notion of morphism between such enhanced sets, obtaining a category MSet containing (a 'copy' of) Set as a full subcategory. (There may be more than one reasonable way to do this! This is intentionally an open-ended exercise.) Which objects in MSet determine ordinary multisets as defined in §2.2 and how? Spell out what a morphism of multisets would be from this point of view. (There are several natural notions of morphisms of multisets. Try to define morphisms in MSet so that the notion you obtain for ordinary multisets captures your intuitive understanding of these objects.)

Problem 11

Draw the relevant diagrams and define composition and identities for the category $C^{A,B}$ mentioned in Example 3.9. Do the same for the category $C^{\alpha,\beta}$ mentioned in Example 3.10.

1.4 Morphisms

Problem 1

Composition is defined for two morphisms. If more than two morphisms are given, e.g.,

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D \xrightarrow{i} E,$$

then one may compose them in several ways, for example:

$$(ih)(gf), (i(hg))f, (i(hg)f), \text{ etc.}$$

so that at every step one is only composing two morphisms. Prove that the result of any such nested composition is independent of the placement of the parenthesis.

Proof. We proceed by induction on n , the number of composable morphisms. The base case $n = 2$ is immediate from the definition of composition. ■

Problem 2

In Example 3.3 we have seen how to construct a category from a set endowed with a relation, provided this latter is reflexive and transitive. For what types of relations is the corresponding category a groupoid.

2 Groups, First Encounter

2.1 Definition of Group

Problem 1

Write a careful proof that every group is the group isomorphism of a groupoid. In particular, every group is the group automorphism of some object of some category.

Problem 2

Consider the 'sets of numbers' listed in sec 1.1 and decide which are made into groups by conventional operations such as $+$ and $\cdot$. Even if the answer is negative, see if variations on the definitions of these sets lead to groups.